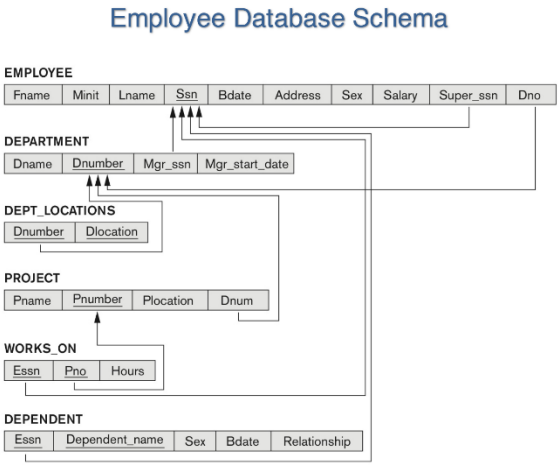
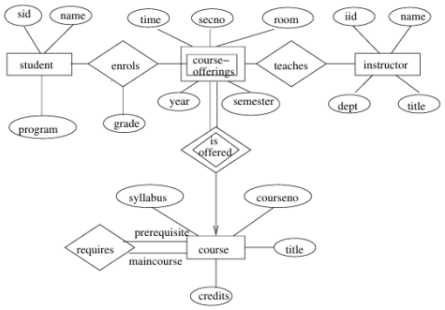
1. Define Data and DBMS
2. What is an IC?
3. What is a multi-valued attribute? Give an example.
4. What are the conditions that should be met to update a View?
5. Give two examples of unary relational operators and explain their significance.
6. What are the conditions for a Union operation to be compatible among two relations?
7. Name and explain at-least two types of outer joins.
8. What are the different quantifiers used in Tuple Relational Calculus and Domain Relational Calculus and discuss their significance?
9. What is the difference between Equijoin and Natural Join?
10. What are DDL and DML? Give one command for each of them.
11. What are the differences between Relational Algebra and Relational Calculus?
12. Define Generalization and Specialization.
13. Discuss the history of DBMS with respect to its evolution
14. What are the advantages of using a DBMS over a file system.
15. What are the levels of data independence? Explain with an example.
16. What are the different components in the structure of a DBMS?
17. What are the schema based Integrity Constraints? Give examples
18. Draw an ERD for the following description:

    We store each employee's name (first, last, MI), Social Security number (SSN), street address, salary, phone (Mobile1, Mobile2), sex (gender), and birth date. An employee is assigned to one department, but may work on several projects, which are not necessarily controlled by the same department. We keep track of the current number of hours per week that an employee works on each project. We also keep track of the direct supervisor of each employee (who is another employee)
19. What are the generic forms of Tuple Relational Calculus and Domain Relational Calculus queries?
20. Given



Write a relational algebraic query to “Find the names of all employees who work on a project located in ‘Hyderabad’”.

21. Given



Identify the weak and strong entities in the ERD.

Construct the appropriate tables for the same.

22. Explain the basic SQL syntax
23. What are the SQL constructs related to the schema based constraints? Explain with an example each.
24. Name any four differences between Tuple relational Calculus and Domain Relational Calculus.
25. Given

EMP_ID	NAME	SALARY	DEPARTMENT
101	Alice	60,000	HR
102	Bob	45,000	IT
103	Charlie	55,000	HR

Write a Tuple Relational Calculus and Domain Relational Calculus query to “Find employees who earn more than 50,000 and also work in the HR department”.

26. Explain the distinctions among the terms primary key, candidate key, and superkey.

27. Identify candidate and primary keys from the following description of a database and justify your choice.

SALESPERSON { string salesperson_code. string employee_number,string name, string contact_info, date hire_date, string department }

CUSTOMER { string customer_number, string tax_id, string email, string name, string contact_info, string address}

ORDER {string order_number, string invoice_number, date order_date, decimal total_amount, string shipping_address }

INVENTORY_ITEM { string item_code, string serial_number, string name, string description, decimal price, int quantity_on_hand}

PART { string part_number, string manufacturer_code, string name, string description, decimal cost, string location}

SUPPLIER { string supplier_code, string vendor_id, string name, string contact_info, string payment_terms}

EMPLOYEE { string employee_id, string social_security, string name, string department, string contact_info, date hire_date }